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REG.NO.: SCT 221-C004-0755/2017
SMA2101 - Calculus 1 CAT

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SMA 2101 CALCULUS I CAT

30th April 2026

1 a) $f(x) = \sqrt{(4-x^2)} = 1 \cdot \sqrt{(x-2)(x+2)}$

let $f(x) = 0 = (x-2)(x+2) = 0$ $\quad x = 2$ or -2

-2	-1	0	1	2
$\sqrt{0}$	$\sqrt{3}$	$\sqrt{4}$	$\sqrt{3}$	$\sqrt{5}$

Domain $= -2 \leq x \leq 2$

Range $= (0, 2)$

b) $\lim_{x \to 2} (x^2-4)/(x-2) = \lim_{x \to 2} \left(\frac{(x-2)(x+2)}{(x-2)}\right) = \lim_{x \to 2}(x+2) = 2+2 = \underline{\underline{4}}$

2 a) $y = (3x^2+2x)(x-5) = x(3x^2+2x) - 5(3x^2+2x) = 3x^3+2x^2-15x^2-10x$

~~$Sy = S\, 9x^2 + 4x - 30x - 10$~~ $= \quad y = 3x^3 - 13x^2 - 10x$

$Sy = S\, 9x^2 - 26x - 10$

$dy/dx = \underline{\underline{9x^2 - 26x - 10}}$

b) tangent of $y = x^3 - 4x$ at $x = 2$

point $\to y(2) = (2)^3 - 4(2) = 0 \Rightarrow (2, 0)$

$dy/dx\; x^3 - 4x = 3x^2 - 4$ $\quad dy/dx \Leftrightarrow m \to$ ~~consta~~ gradient

~~$\therefore y = mx + c$~~ replace in $3x^2 - 4 \Rightarrow 3(2)^2 - 4 \Rightarrow 8$

but ~~$y = mx + c$~~ $dy/dx = m$ $\Leftrightarrow$ ~~replace~~ $dy = m dx$

$y - 0 = m(x-2) \Rightarrow y = m(x-2)$ but $m = 8$

$y = 8(x-2) = 8x - 16$

$y = \underline{\underline{8x - 16}}$

3 a) $\int (3x^2 - 2x + 1)\, dx$

$\int 3x^2\, dx = x^3$

$\int -2x\, dx = -x^2$

$\int 1\, dx = x + c$

$\int (3x^2 - 2x + 1)\, dx = x^3 - x^2 + x + c$

b) area of curve $y = x^2$ at $x = 0$ and $x = 3$

~~x~~	~~0~~	~~3~~
~~y~~	~~0~~	~~9~~

but area $= \int_3^0 x^2\, dx = \left[\frac{x^3}{3}\right] = \frac{3^3}{3}$

$= \frac{27}{3} = \underline{\underline{9}}$